1. SIGNAL GENERATION AND BASIC SIGNAL OPERATIONS

Experiment 1.1

Generation of Continuous Time Signals

Aim

To verify the generation of a sinusoidal signal with a specific frequency, amplitude, and phase shift using MATLAB. The Aim is to ensure that the signal generator can accurately produce a sinusoidal waveform with the desired parameters.

Apparatus Required

PC with MATLAB.

Program:

```
% Generation of Continuous Time Signals

clc;

clear;

close all;

% Time axis

t = -5:0.01:5;

% Continuous-time sine signal

x = sin(2*pi*t);

% Plotting

figure;

plot(t, x, 'LineWidth', 2);

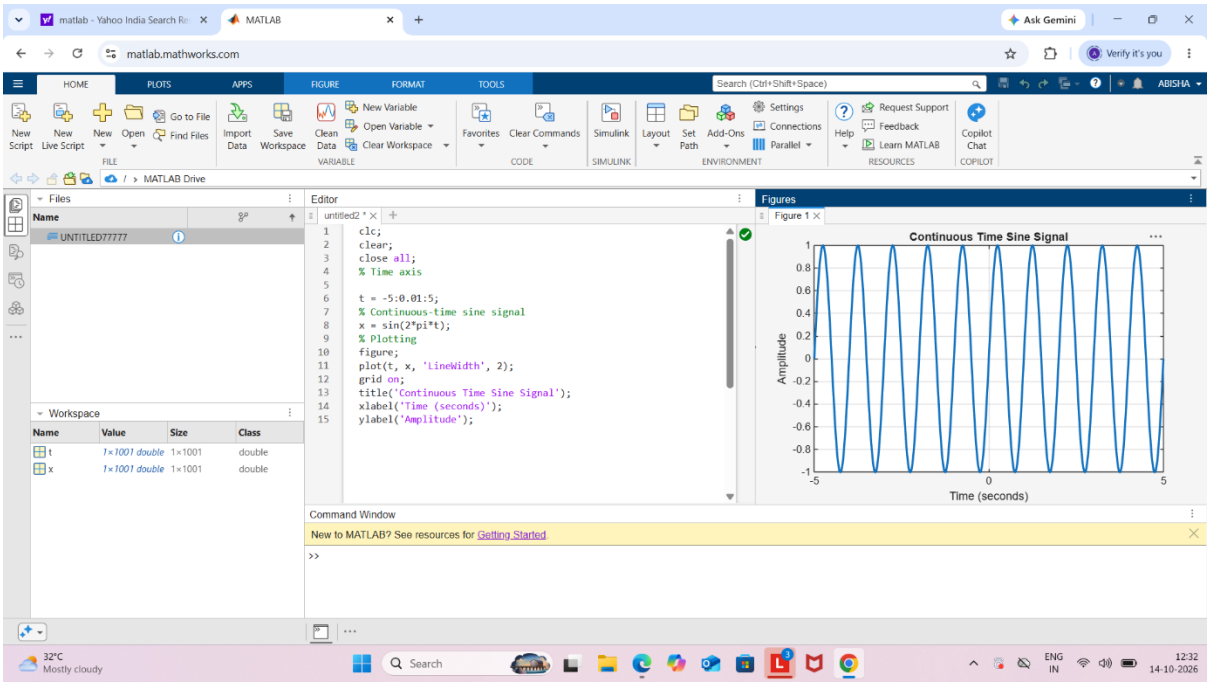
grid on;

title('Continuous Time Sine Signal');

xlabel('Time (seconds)');

ylabel('Amplitude');
```

Output Waveform:



Result:

This test case validates that the signal generator can produce a sinusoidal waveform with the specified frequency, amplitude, and phase shift.

Experiment 1.2

Generation of Discrete Time Signals

Aim:

To generate and verify a discrete-time unit impulse signal using MATLAB. The Aim is to ensure that the signal generator can accurately produce an impulse signal at a specified time index, which is crucial for analyzing and understanding the behavior of discrete-time systems.

Apparatus Required

PC with MATLAB.

Program:

```
% Generation of Discrete Time Signal

clc;

clear;

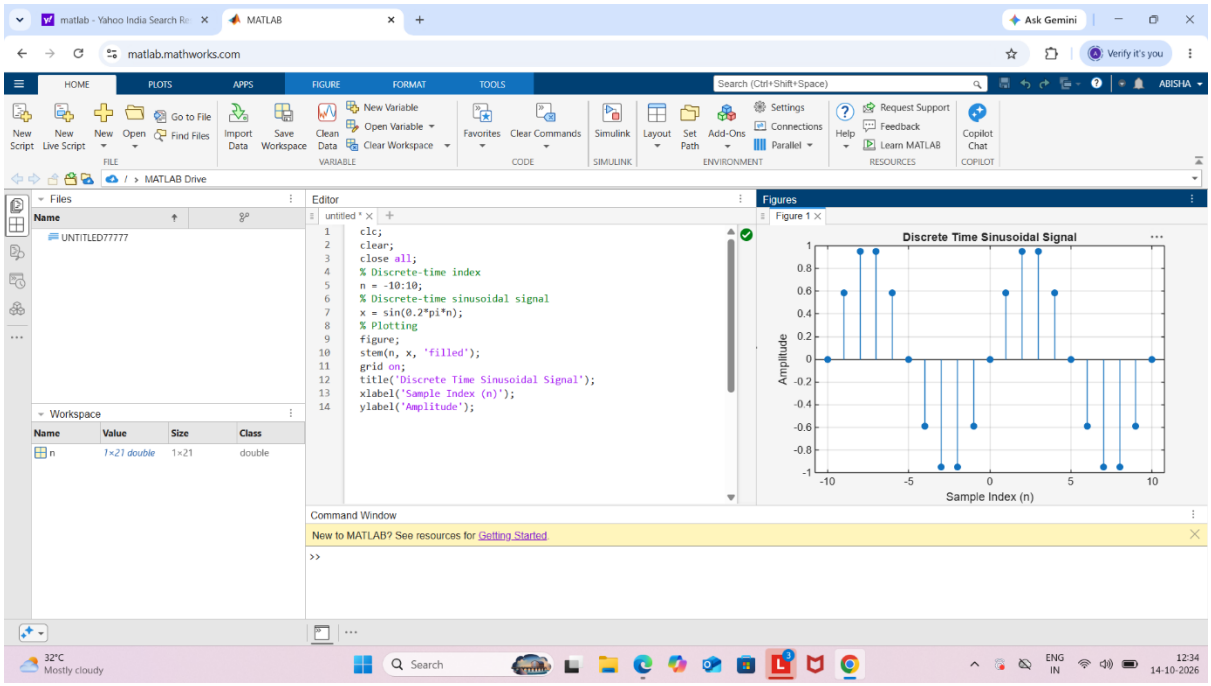
close all;

% Discrete-time index

n = -10:10;
```

```
% Discrete-time sinusoidal signal
x = sin(0.2*pi*n);
% Plotting
figure;
stem(n, x, 'filled');
grid on;
title('Discrete Time Sinusoidal Signal');
xlabel('Sample Index (n)');
ylabel('Amplitude');
```

Output Waveform:



Result:

This test case validates that the signal generator can accurately produce a discrete-time unit impulse signal at a specified index, which is essential for analyzing the response of discrete-time systems.

Experiment 1.3

Generation of Triangular and Sawtooth Signals

Aim

Generate and plot triangular and sawtooth signals.

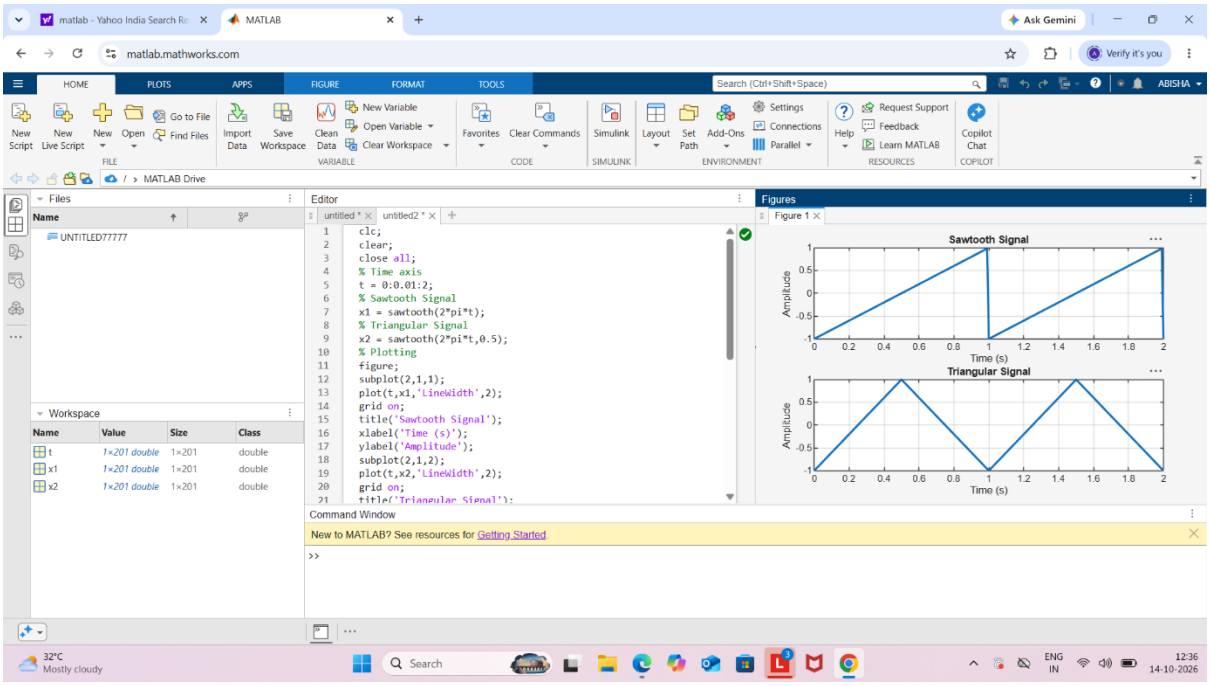
Apparatus Required

PC with MATLAB.

Program (MATLAB)

```
% Generation of Triangular and Sawtooth Signals
clc;
clear;
close all;
% Time axis
t = 0:0.01:2;
% Sawtooth Signal
x1 = sawtooth(2*pi*t);
% Triangular Signal
x2 = sawtooth(2*pi*t,0.5);
% Plotting
figure;
subplot(2,1,1);
plot(t,x1,'LineWidth',2);
grid on;
title('Sawtooth Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(2,1,2);
plot(t,x2,'LineWidth',2);
grid on;
title('Triangular Signal');
xlabel('Time (s)');
ylabel('Amplitude');
```

Output Generated Graph



Result

Triangular wave has equal rise and fall; sawtooth shows sharp drop and gradual rise.

Experiment 1.4

Step and Ramp Functions (Continuous & Discrete)

Aim

Generate and plot unit impulse, unit step, and ramp functions.

Apparatus Required

PC with MATLAB.

Program (MATLAB)

```
% Step and Ramp Functions (Continuous & Discrete)

clc;

clear;

close all;

%% Continuous-Time Signals

t = -5:0.01:5;

% Continuous Unit Step Function

u = double(t >= 0);

% Continuous Ramp Function
```

```
r = t .* (t >= 0);
%% Discrete-Time Signals
n = -5:5;
% Discrete Unit Step Function
ud = double(n >= 0);
% Discrete Ramp Function
rd = n .* (n >= 0);
%% Plotting
figure;
subplot(2,2,1);
plot(t,u,'LineWidth',2);
grid on;
title('Continuous Unit Step Function');
xlabel('Time (t)');
ylabel('Amplitude');
subplot(2,2,2);
plot(t,r,'LineWidth',2);
grid on;
title('Continuous Ramp Function');
xlabel('Time (t)');
ylabel('Amplitude');
subplot(2,2,3);
stem(n,ud,'filled');
grid on;
title('Discrete Unit Step Function');
xlabel('Sample Index (n)');
ylabel('Amplitude');
subplot(2,2,4);
stem(n,rd,'filled');
grid on;
title('Discrete Ramp Function');
xlabel('Sample Index (n)');
ylabel('Amplitude');
```

Output Generated Graph

